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Cyclistic Bike-Share Case Study

Differences in Cyclistic Bike Usage Between Annual Members and Casual Riders

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Cyclistic Bike-Share Case Study

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Executive Summary

This case study analyzes 5,951,516 Cyclistic bike trips recorded from September 2025 through August 2026 to identify differences between annual members and casual riders. Annual members accounted for 65.04% of all rides, while casual riders represented 34.96%. Casual riders took longer trips, averaging 18.41 minutes compared with 12.22 minutes for members. Members showed strong usage during typical commuting hours, particularly around 8:00 a.m. and 5:00 p.m., while casual ridership was more concentrated during afternoons, weekends, and warmer months. Electric bikes were the most frequently used bike type for both groups. These findings suggest that Cyclistic should target frequent casual riders with membership campaigns emphasizing cost savings, convenience, electric-bike access, and the value of membership for repeated weekend and recreational use.

Business Task

Cyclistic wants to increase the number of casual riders who purchase annual memberships. The purpose of this analysis is to determine how annual members and casual riders use Cyclistic bikes differently and use those findings to support a data-driven marketing strategy.

The primary business question is:

How do annual members and casual riders use Cyclistic bikes differently?

The analysis also supports the following marketing considerations:

1. Why might casual riders purchase an annual membership?
2. How could Cyclistic use digital media to influence casual riders?
3. Which customer behavior should Cyclistic target in a membership campaign?

Key Stakeholders

- **Lily Moreno:** Director of Marketing and manager responsible for the membership-growth campaign.
- **Cyclistic Marketing Analytics Team:** Responsible for collecting, analyzing, and reporting customer-usage trends.
- **Cyclistic Executive Team:** Responsible for reviewing and approving the recommended marketing strategy.
- **Casual Riders:** The customer group targeted for conversion to annual memberships.

Data Source

The analysis used 12 months of Cyclistic historical trip data covering September 2025 through August 2026. The data was provided in 12 monthly CSV files and contained 6,115,982 original trip records.

The dataset included the following relevant fields:

- Ride identification number
- Bike type
- Ride start and end timestamps
- Starting and ending station names

-
- Starting and ending coordinates
 - Rider type: annual member or casual rider

Cyclistic's trip data does not contain personally identifiable rider information. It also does not explain each rider's purpose, motivation, income, or reason for selecting a particular membership type. Therefore, conclusions about rider motivation are presented as data-supported interpretations rather than confirmed customer intentions.

Tools Used

- **Microsoft Excel and Power Query:** Initial data preparation, quality checks, pivot tables, and exploratory analysis.
- **Google BigQuery and SQL:** Large-scale data cleaning, validation, summary views, and analytical queries.
- **Looker Studio:** Interactive BigQuery reporting and visual validation.
- **Python and JupyterLab:** Independent data cleaning, validation, aggregation, chart creation, and export of Tableau-ready datasets.
- **Tableau Public:** Interactive dashboards, rider-behavior comparisons, station maps, and portfolio publication.

Using multiple tools provided an opportunity to cross-check the results. The final Python dataset contained 5,951,516 valid trips, matching the clean total established during the analysis.

Data Preparation and Cleaning

The 12 monthly CSV files were combined into a single dataset. The original combined data contained 6,115,982 rows. The cleaning process applied consistent rules across Excel, BigQuery, and Python.

The following steps were performed:

1. Combined all monthly files covering September 2025 through August 2026.
2. Standardized the start and end timestamp fields to account for differences in date and time formatting.
3. Removed records with missing ride IDs, start times, end times, or rider classifications.
4. Limited rider classifications to members and casual.
5. Removed duplicate records based on the ride ID.
6. Calculated ride length in minutes using the difference between the ending and starting timestamps.
7. Removed rides shorter than one minute or longer than 1,440 minutes.
8. Limited the dataset to rides starting on or after September 1, 2025, and before September 1, 2026.
9. Created additional fields for the day number, weekday name, ride month, and starting hour.
10. Validated the final dataset for missing required values, duplicate ride IDs, invalid rider types, invalid ride lengths, and trips outside the analysis period.

Data-Quality Results

- Original rows: 6,115,982
- Clean rows: 5,951,516
- Rows removed: 164,466
- Percentage removed: 2.69%
- Remaining missing required values: 0

- Remaining duplicate ride IDs: 0
- Remaining invalid rider types: 0
- Remaining invalid ride lengths: 0
- Remaining trips outside the analysis period: 0

The cleaned data was saved as a Parquet file to support efficient reuse in Python. Smaller aggregated CSV files were also created for the Tableau dashboards.

Analysis and Key Findings

1. Members Accounted for Most Rides

Annual members completed 3,870,680 rides, representing 65.04% of all clean trips. Casual riders completed 2,080,836 rides, representing 34.96%. Members therefore generated almost twice as many rides as casual users during the analysis period.

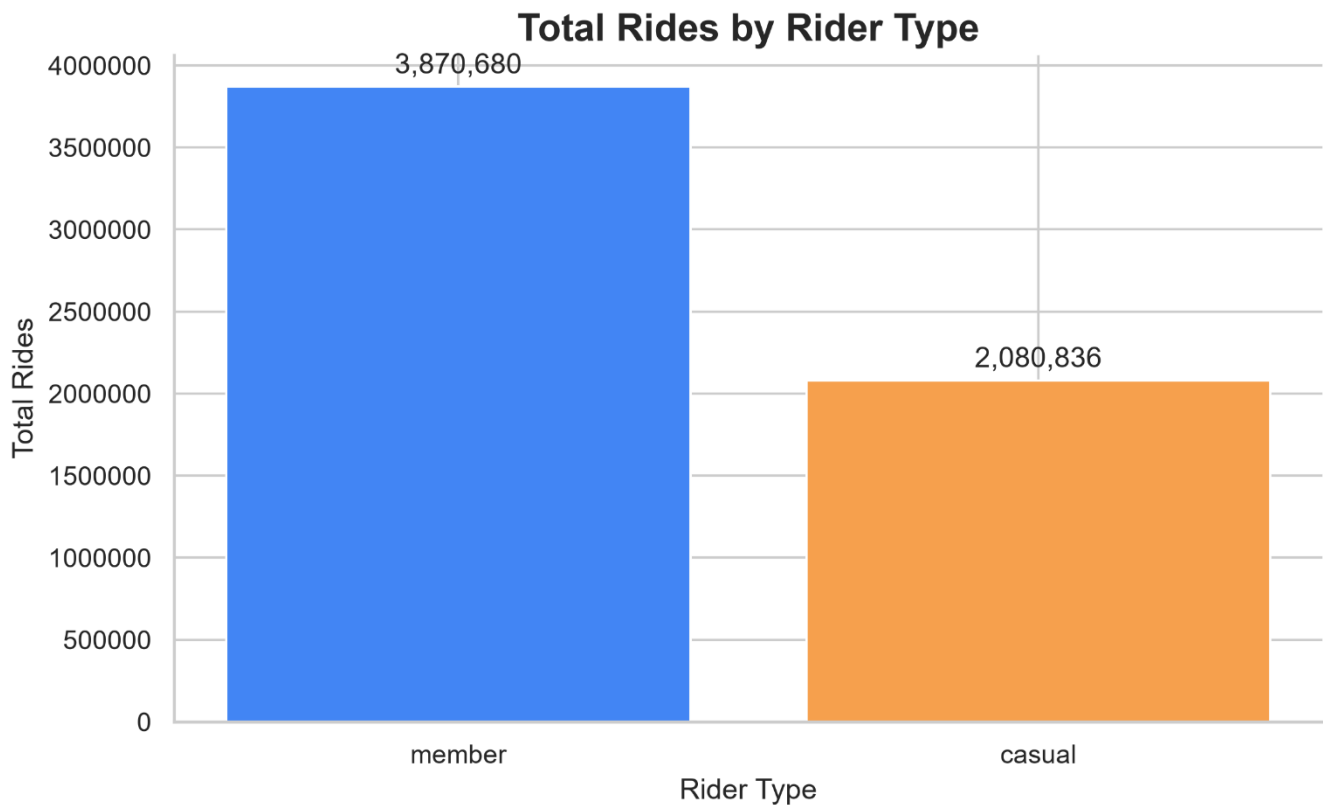


Figure 1. Total rides completed by casual riders and annual members.

2. Casual Riders Took Longer Trips

Casual rides averaged 18.41 minutes, compared with 12.22 minutes for members. Although members rode more frequently, casual riders remained on the bikes approximately six minutes longer per trip. This difference suggests that casual riders are more likely to use the service for leisure, sightseeing, or other less time-sensitive activities.

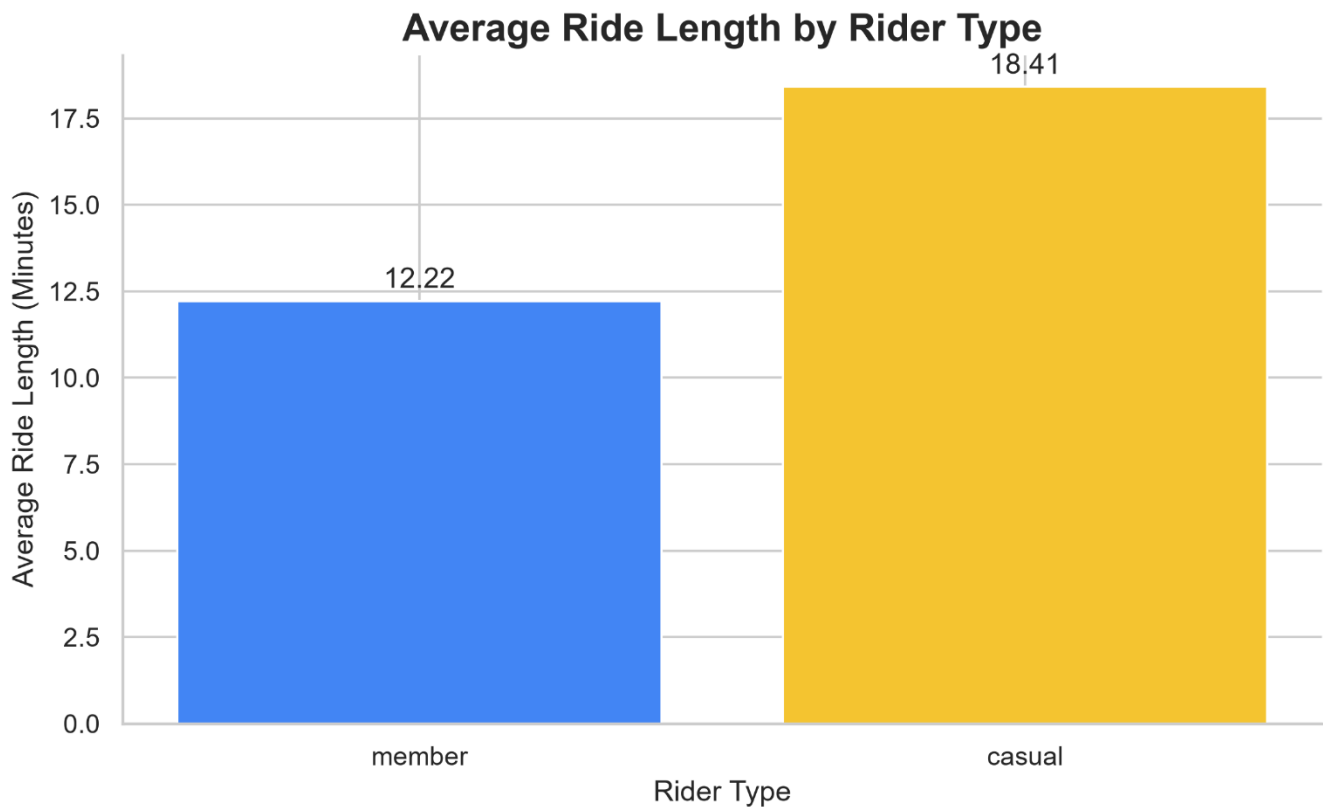


Figure 2. Average ride length in minutes by rider type.

3. Members Displayed More Routine Weekday Behavior

Member ridership remained strong from Monday through Friday and reached its highest weekday level on Wednesday, with 618,853 rides. Casual usage was lower during the workweek and increased substantially at weekends. Saturday was the most popular day overall, with 929,681 combined rides, including 438,505 casual rides.

These patterns suggest that members use Cyclistic more regularly for transportation, while casual riders are more strongly associated with weekend and recreational activity.

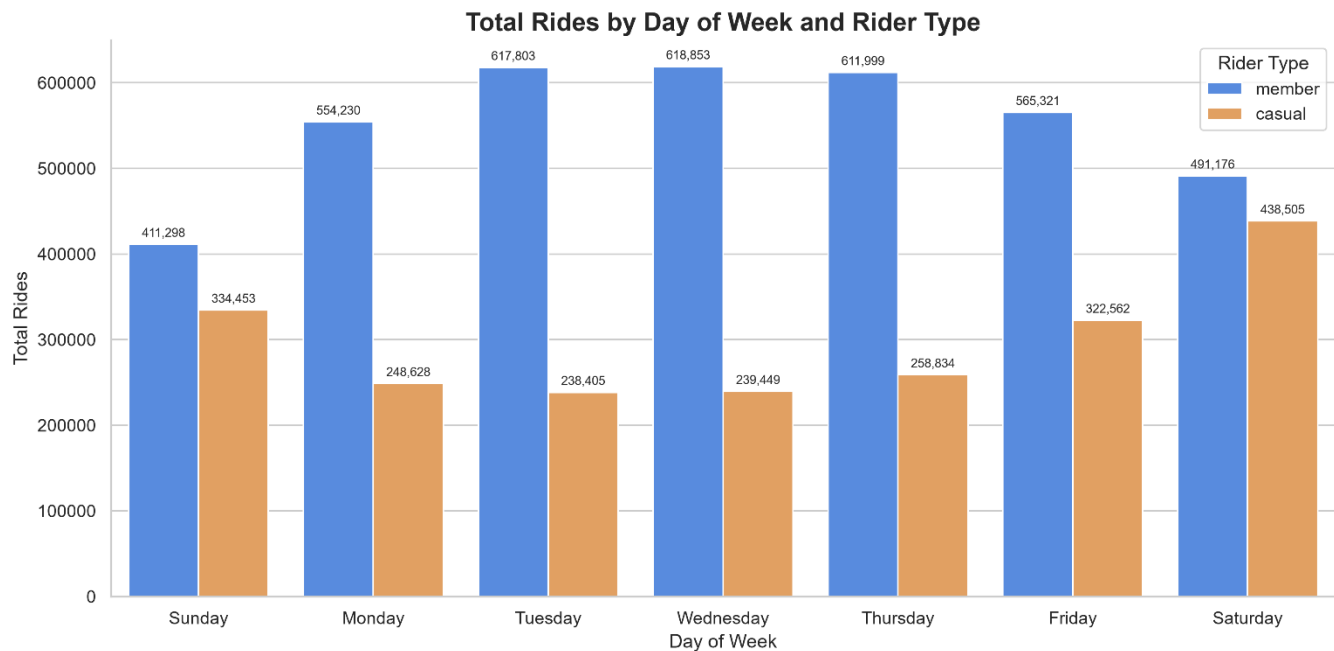


Figure 3. Total rides by weekday and rider type.

4. Members Showed Clear Commuting-Hour Peaks

Member ridership rose sharply around 8:00 a.m. and reached its highest hourly level at approximately 5:00 p.m., with 417,894 rides. A second major member peak occurred at 4:00 p.m., with 366,416 rides. These morning and afternoon increases are consistent with commuting behavior.

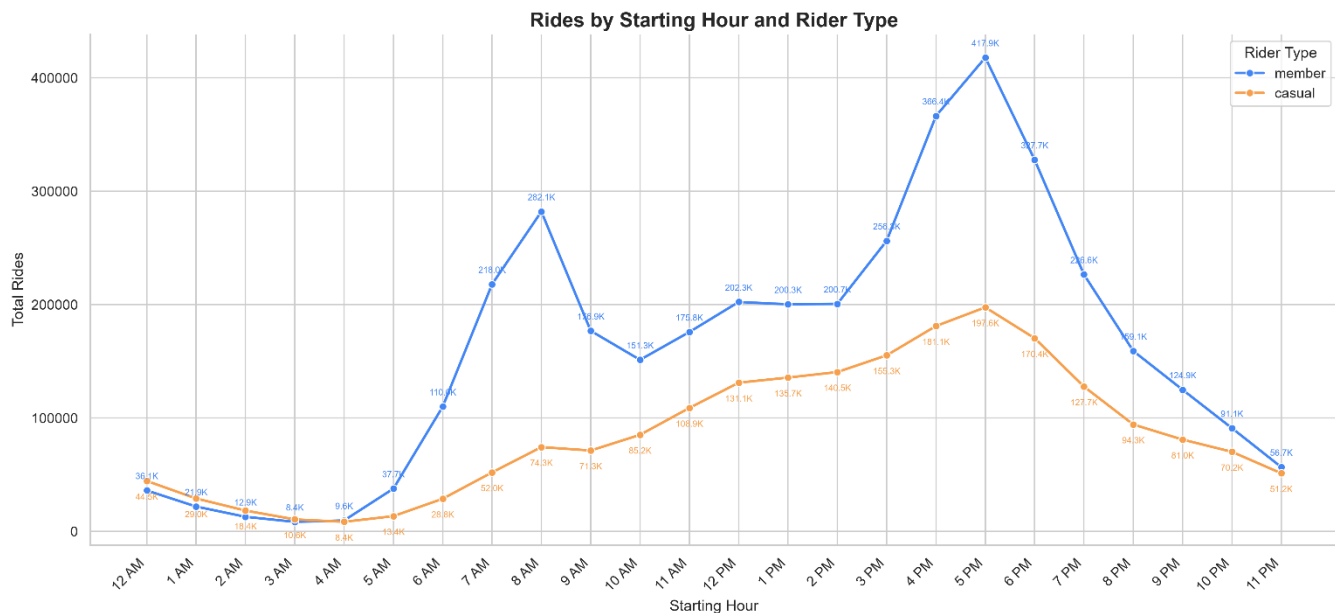


Figure 4. Total rides by starting hour and rider type.

Casual ridership increased more gradually during the day and peaked around 5:00 p.m., with 197,600 rides. Unlike members, casual riders did not display a comparably strong morning peak.

5. Ridership Was Seasonal

Usage declined considerably during the winter and increased during spring and summer. January 2026 was the lowest-volume month, with approximately 133,743 combined rides. July and August 2026 were the highest-volume months, each recording approximately 846,000 combined rides.

Seasonal growth was especially important among casual riders, indicating that warmer months provide the strongest opportunity for membership-conversion campaigns.

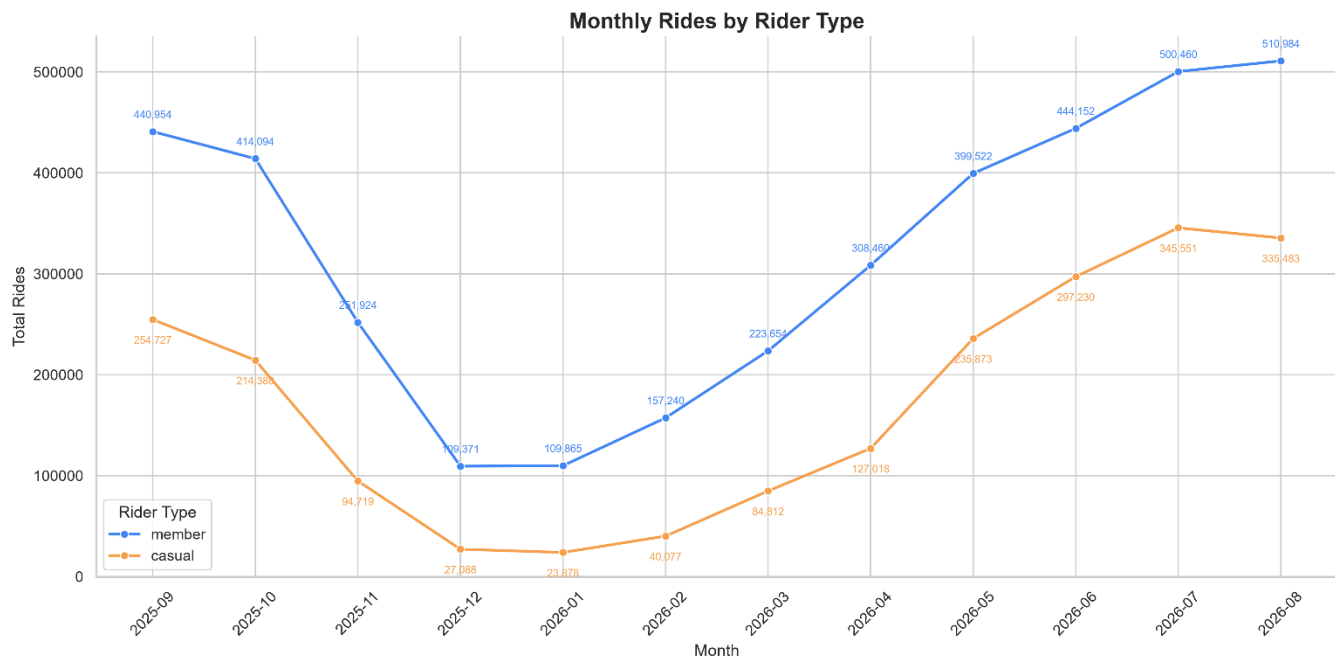


Figure 5. Monthly ride trends by rider type.

6. Electric Bikes Were Preferred by Both Groups

Members completed 2,608,238 electric-bike rides and 1,262,442 classic-bike rides. Casual riders completed 1,514,865 electric-bike rides and 565,971 classic-bike rides. Electric bikes accounted for approximately 67% of member rides and 73% of casual rides.

This makes electric bike access a strong feature to emphasize in campaigns directed toward casual riders.

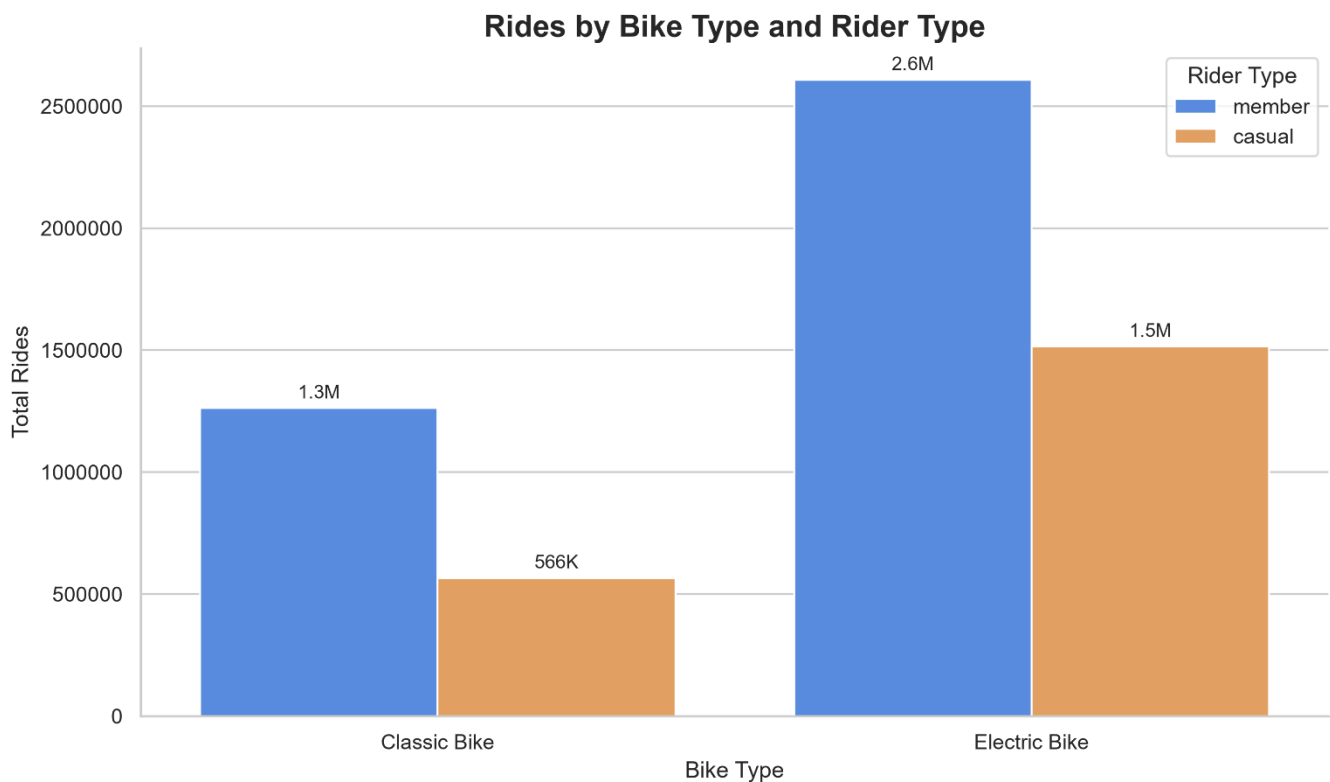


Figure 6. Bike-type usage by rider type.

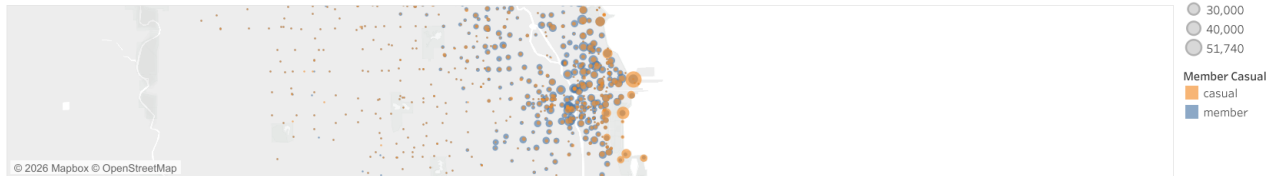
7. Popular Stations Were Strongly Associated with Casual Use

The station analysis showed that Navy Pier was the busiest starting and ending station among the stations included in the geographic dataset. It recorded 68,030 starting rides and 67,327 ending rides. Most of those rides were completed by casual riders.

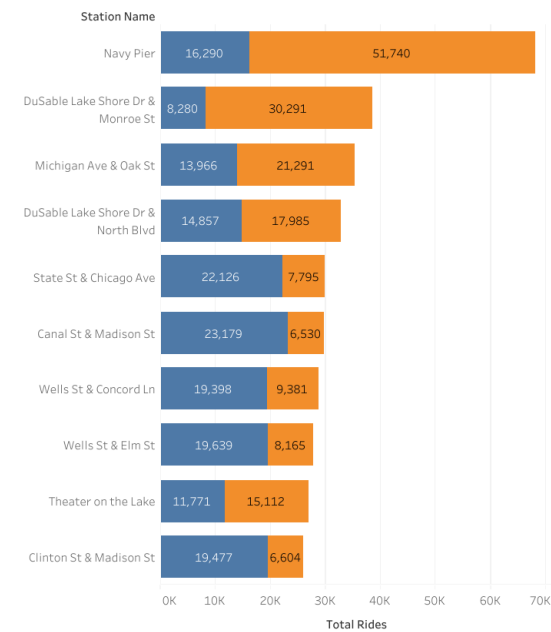
Other high-volume locations included DuSable Lake Shore Drive and Monroe Street, Michigan Avenue and Oak Street, and DuSable Lake Shore Drive and North Boulevard. These destinations are concentrated near recreational, waterfront, and visitor-oriented areas, making them valuable locations for targeted membership messaging.

Station Analysis

Station Popularity Map



Top Starting Stations



Top Ending Stations

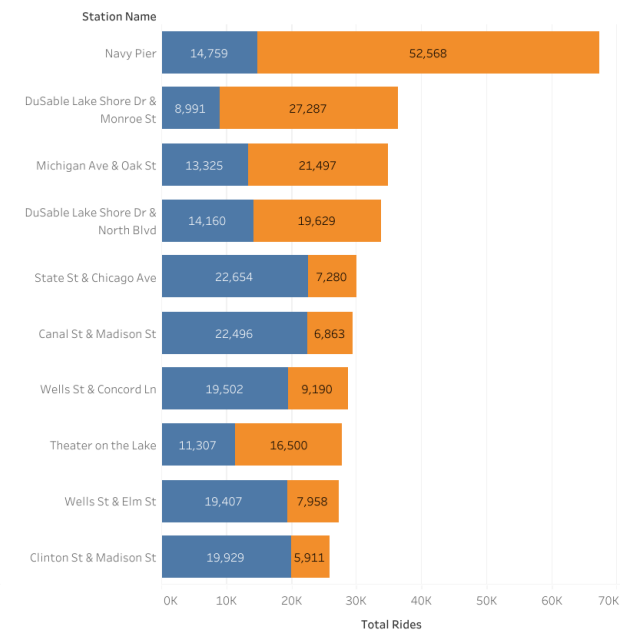


Figure 7. Cyclistic station popularity and the ten highest-volume starting and ending stations.

Answer to the Primary Business Question

Annual members use Cyclistic more frequently and demonstrate patterns consistent with routine transportation, including strong weekday activity and pronounced commuting-hour peaks. Casual riders take longer trips and are more active on weekends, during warmer months, in the afternoon, and around recreational destinations. Both groups prefer electric bikes, but the preference is even stronger among casual riders.

Marketing Recommendations

1. Target Frequent Casual Riders with Personalized Membership Offers

Cyclistic should identify casual riders who make repeated trips and present them with personalized comparisons showing how much they could save with annual membership.

The campaign should emphasize predictable pricing, convenience, and the value of unlimited or discounted repeat use.

Promotions could be delivered through the Cyclistic mobile app, email, and post-ride notifications. Messaging such as “You have taken multiple rides this month—see how much membership could save you” would connect the offer directly to the rider’s actual behavior.

2. Launch Seasonal Weekend Campaigns

Casual ridership is strongest during weekends and warmer months, particularly from late spring through summer. Cyclistic should launch conversion campaigns before and during this high-demand period.

Digital advertisements should focus on social media, mobile search, tourism platforms, and local event websites. Weekend-specific trial offers could allow casual riders to apply part of their recent ride spending toward annual membership. Campaign content should emphasize that membership is useful for recreation as well as commuting.

3. Promote Membership at High-Traffic Stations and Emphasize Electric Bikes

Cyclistic should concentrate digital and location-based promotions around high-volume stations such as Navy Pier and the DuSable Lake Shore Drive locations. QR codes, in-app

notifications, geofenced advertisements, and station signage could reach casual riders while they are actively using the service.

Because approximately 73% of casual rides use electric bikes, promotional messages should prominently feature electric-bike access. Campaigns could show how an annual membership makes frequent electric-bike use more convenient and cost-effective.

Suggested Campaign Measures

Cyclistic should monitor the following measures to determine whether the campaigns are effective:

- Casual-to-member conversion rate
- Membership purchases by campaign and station
- Promotional link and QR-code engagement
- Conversion rate among repeat casual riders
- Cost per membership conversion
- Member retention after 30, 90, and 365 days
- Changes in weekend and seasonal membership usage

A controlled campaign test should compare targeted offers, messaging, stations, and digital channels before the strategy is expanded across the entire system.

Limitations and Opportunities for Further Analysis

The available trip data describes rider behavior but does not contain demographic information, membership pricing history, marketing exposure, customer satisfaction, or the reason for each trip. Therefore, the analysis can identify usage patterns but cannot directly establish why casual riders choose not to purchase memberships.

Trips without valid station names or coordinates could not be included in the station-level analysis. The station charts therefore represent rides with sufficient geographic information and may not include every ride in the cleaned dataset.

Future analysis should be incorporated:

- Customer surveys about membership barriers and riding motivations
- Membership pricing and promotional history
- Weather conditions
- Local events and tourism patterns
- Repeat-rider frequency
- App engagement and campaign-response data
- Conversion and retention results from controlled marketing tests

Conclusion

Annual members and casual riders demonstrate clearly different usage patterns. Members account for most rides and show consistent weekday and commuting-hour activity. Casual riders take longer trips and ride more frequently on weekends, during warmer months, in the afternoon, and near recreational destinations.

Cyclistic can use these differences to develop a more focused conversion strategy. The strongest opportunities are to target frequent casual riders with personalized savings messages, launch seasonal weekend campaigns, and promote membership near popular recreational stations while emphasizing electric-bike access.

These recommendations should be tested through controlled digital campaigns and customer research. Combining behavioral trip data with direct customer feedback would allow Cyclistic to better understand membership barriers and improve future conversion efforts.

Interactive Dashboards and Supporting Analysis

The interactive dashboards provide additional detail and allow users to explore rider behavior across time, rider type, bike type, and station location.

- **Tableau Public – Cyclistic Bike-Share Analysis:** [[Tableau Public link](#)]
- **Looker Studio – Cyclistic BigQuery Analysis:**

<https://lookerstudio.google.com/reporting/c46d3db8-7ef6-46ab-8e1c-5d352520da02>